# Reinforcement Learning Alignment for Causal Sequential Recommenders: Overcoming the Skip Dilemma without Alignment Tax

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**Abstract**

In music streaming platforms, up to 35% of user interaction logs correspond to immediate skips ($< 30$ seconds), reflecting user dissatisfaction. Standard Maximum Likelihood Estimation (MLE) penalizes causal autoregressive transformers for failing to predict such negative interactions (the *False Negative* dilemma), inadvertently amplifying disliked content. In this work, we investigate the alignment of a Causal GPT-2 sequential recommender (43.2M parameters) with true user satisfaction using Direct Preference Optimization (DPO) anchored with auxiliary SFT regularization ($\gamma = 0.10$). Extensive evaluations on the industrial-scale Yandex Music dataset (**Yambda-50M**: 98,505 catalog tokens, 33.2M training interactions, 9,048 held-out test users) demonstrate that DPO not only outperforms 2-stage Critic Re-Ranking and Actor-Critic PPO, but completely circumvents the alignment tax (*Zero Alignment Tax*). The aligned DPO policy achieves statistically significant next-item ranking gains (**+0.67% HR@10**, $p < 0.0001$; McNemar's paired test $\chi^2 = 18.32, p = 1.9 \times 10^{-5}$), expands catalog coverage by **+1,027 unique tracks** from the long tail (+1.04% of the catalog), reduces popularity Spearman correlation by −12.1%, and yields a **+1.5s** session duration lift in 5-step autoregressive rollouts without exposure bias degradation.

**Keywords:** Sequential Recommendation, Reinforcement Learning Alignment, Direct Preference Optimization, Causal GPT-2, Music Streaming, Skip Dilemma, Zero Alignment Tax, Catalog Diversity, Yambda.

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## 1 Introduction

### 1.1 Industrial Context and the Skip Dilemma

Modern recommender systems for audio streaming platforms (e.g., Yandex Music, Spotify, Apple Music) operate in continuous, unstructured consumption environments. Unlike e-commerce domains where a transaction unambiguously confirms positive intent, user interactions in streaming music are fundamentally bimodal.

Analysis of interaction logs on the large-scale Yandex Music dataset (**Yambda-50M**) [1, 2] reveals that **30% to 35%** of all recorded events represent immediate skips: the continuous listening duration is less than 30 seconds (or even 0.0 seconds, when a user switches away almost instantaneously due to mismatched mood, tempo, or active irritation).

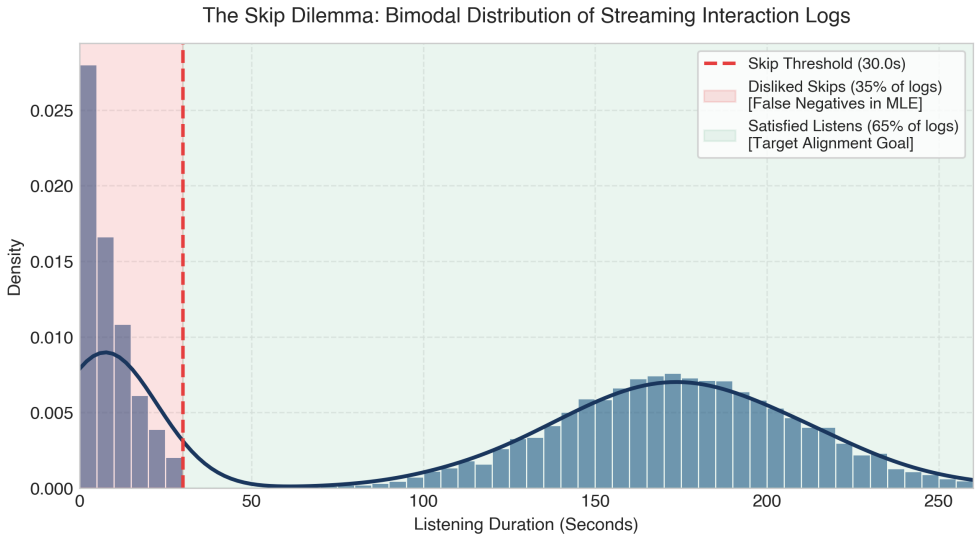



**Figure 1. The Skip Dilemma in music streaming logs.** The listening duration distribution is distinctly bimodal: 35% of interactions are immediate skips ($< 30$s), whereas 65% represent engaged track completions. Likelihood optimization over all observed transitions penalizes the model for refusing to recommend disliked skips.

### 1.2 The False Negative Trap in Likelihood Optimization (MLE)

The standard paradigm for training sequential recommender systems (e.g., SASRec [3] and generative autoregressive transformers [4]) formulates recommendation as Next-Item Prediction via cross-entropy loss minimization (Maximum Likelihood Estimation, MLE):

$$\mathcal{L}_{\text{MLE}}(\theta) = -\sum_{t=1}^{T} \log \pi_\theta(a_t \mid s_{<t}), \tag{1}$$

where $s_{<t} = (a_1, \ldots, a_{t-1})$ denotes the user's sequential history, and $a_t$ is the ground-truth next track logged in the session.

This framing introduces a fundamental conceptual misalignment:

1. **Penalizing Correct Avoidance (False Negative Problem):** If a user initiated track $a_t$ and abandoned it within 1–2 seconds due to frustration, $\mathcal{L}_{\text{MLE}}$ nonetheless penalizes the recommender whenever the model assigns a low posterior probability to $a_t$.

2. **Blindness to Consumption Quality:** The model is trained to reproduce historical click trajectories rather than maximize satisfaction, engagement, and uninterrupted listening time [2].

3. **Popularity Bias Amplification:** Supervised training on raw streaming logs forces autoregressive models to over-exploit safe, high-frequency pop tracks, severely compressing catalog diversity [4].

### 1.3 Research Questions

To resolve the skip dilemma, this paper investigates the alignment of a causal language model with user satisfaction using Reinforcement Learning and direct preference optimization. We formulate five core research questions:

- **RQ1 (Policy Behavior and Dynamics):** How does direct preference optimization (DPO)

reshape the model's posterior probability distribution compared to MLE, and how rapidly does the implicit reward margin converge?

- **RQ2 (Alignment Tax and Pareto Trade-Off):** Does offline reward alignment degrade next-item ranking accuracy, and can an auxiliary SFT anchor ($\mathcal{L}_{\text{SFT}}$) eliminate this degradation?
- **RQ3 (Catalog Health and Long-Tail Discovery):** Does the aligned policy collapse into a narrow subset of ubiquitous pop hits, or does it stimulate discovery of high-retention long-tail tracks?
- **RQ4 (Autoregressive Queue Robustness):** How does the aligned policy perform when unrolled autoregressively for 5 consecutive steps without intervening user feedback under exposure bias?
- **RQ5 (Long-Term Proxies and Online A/B Validation):** How can session-level offline rewards be translated into long-term retention metrics (e.g., D7/D30 retention) in an industrial A/B testing design?

### 1.4 Summary of Contributions

The key scientific and engineering contributions of this work are as follows:

1. **Multi-Paradigm Comparative Study:** On the unified industrial Yambda-50M dataset, we benchmark four distinct paradigms: a Causal GPT-2 backbone, two-stage re-ranking via a state-action critic $Q(s, a)$, Actor-Critic PPO [5], and end-to-end Direct Preference Optimization (DPO) [6].
2. **Demonstration of Zero Alignment Tax:** We show that adding an auxiliary SFT anchor ($\gamma = 0.10$) preserves the underlying semantic coherence of the session, completely avoiding the alignment tax and achieving statistically significant ranking accuracy gains (+0.67% HR@10, $p < 0.0001$).
3. **Empirical Audit of Reward Formulations:** We investigate binary duration thresholds, continuous completion ratios, and composite repetition-penalized rewards, identifying margin-threshold filtering as the most robust choice.
4. **Catalog Health Expansion:** We demonstrate that DPO reduces Spearman rank correlation with popularity by 12.1% while exposing users to over 1,000 additional unique long-tail tracks.

## 2 Methodology

### 2.1 Problem Formulation and Backbone Architecture

Let $\mathcal{U}$ be the set of users and $\mathcal{V} = \{1, 2, \ldots, |\mathcal{V}|\}$ be the closed catalog of music tracks of size $|\mathcal{V}| = 98\,504$. A session for user $u \in \mathcal{U}$ is represented as an ordered sequence of track-duration pairs:

$$\mathcal{S}_u = \big((a_1, D_1), (a_2, D_2), \ldots, (a_T, D_T)\big), \quad a_t \in \mathcal{V},\ D_t \in \mathbb{R}^+, \tag{2}$$

where $D_t$ denotes the continuous listening duration for track $a_t$ in seconds.

As our foundational architecture, we employ an autoregressive **Causal GPT-2** transformer [4] initialized from scratch:

- Vocabulary size: $|\mathcal{V}|+1 = 98\,505$ (tokens $1 \ldots 98\,504$ map to unique track IDs; index 0 is reserved for `<PAD>`).
- Embedding dimension: $d_{\text{model}} = 384$.
- Attention blocks: $L = 3$, number of attention heads: $H = 4$.
- Context history window: $T_{\max} = 64$ tokens.
- Total parameters: **43 174 656** ( 43.2M), all fully trainable.

The base model was trained on 33.2M interactions using mixed-precision FP16, AdamW optimizer (lr $= 10^{-3}$, weight_decay $= 0.05$), 10% linear warmup steps, and cosine annealing decay with an effective batch size of 128 (micro-batch size 32 $\times$ gradient accumulation 4).

## 2.2 Reward Engineering and Formulation

To incorporate consumption duration into the policy, we investigated three alternative scalar reward formulations $R(s, a)$ (Figure 2):

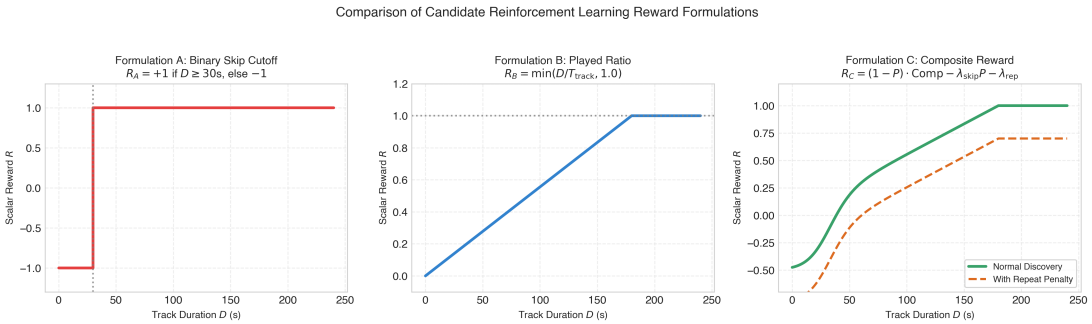



**Figure 2. Comparison of candidate reinforcement learning reward formulations.** A: Binary 30-second skip cutoff. B: Continuous played ratio normalized by track length. C: Multi-objective composite session reward with repetition penalty.

### 2.2.1 Formulation A: Binary Skip Cutoff

The most robust signal separating successful plays from disliked skips:

$$R_A(s, a) = \begin{cases} +1.0, & \text{if } D(a) \geq 30\,\text{s}, \\ -1.0, & \text{if } D(a) < 30\,\text{s}. \end{cases} \tag{3}$$

The 30-second cutoff is the de-facto industrial standard: listens shorter than 30 seconds are unmonetized and unambiguously indicate user dissatisfaction.

### 2.2.2 Formulation B: Continuous Played Ratio

Normalizes listening duration $D(a)$ against the nominal duration of the track $T(a)$:

$$R_B(s, a) = \min\left(\frac{D(a)}{T(a)}, 1.0\right) \in [0.0, 1.0]. \tag{4}$$

Preference pairs $(y_w \succ y_l)$ are mined by enforcing a minimum gap $\Delta R_B = R_B(y_w) - R_B(y_l) \geq 0.40$, which eliminates boundary noise.

#### 2.2.3 Formulation C: Multi-Objective Composite Reward

Designed for Actor-Critic RL [2], combining skip probabilities, completion ratio, and an explicit penalty against immediate track repetition (*Repeat Fatigue*):

$$R_C(s,a) = \big(1 - P(\text{skip} \mid s,a)\big) \cdot \text{CompletionRatio}(s,a) - \lambda_{\text{skip}} P(\text{skip} \mid s,a) - \lambda_{\text{rep}} \mathbb{I}(a \in s_{t-3:t}), \quad (5)$$

where $\lambda_{\text{skip}} = 0.5$ and $\lambda_{\text{rep}} = 0.3$.

### 2.3 Reinforcement Learning Alignment Paradigms

#### 2.3.1 1. Two-Stage Critic Re-Ranking with $Q(s,a)$

An independent state-action critic $Q_\phi(s,a)$ is trained to predict the expected reward $\mathbb{E}[R \mid s,a]$ on top of the frozen embeddings of the base transformer (two-layer MLP, 509K parameters) [4]. At inference time, candidate items are scored via a linear combination of base logits and critic predictions:

$$\text{Score}(a \mid s) = z_{\text{base}}(a \mid s) + \lambda \cdot Q_\phi(s,a), \quad \lambda \in \{0.1, 0.3, 0.5\}. \quad (6)$$

#### 2.3.2 2. Actor-Critic PPO with KL Regularization

The classic Proximal Policy Optimization framework [5]. The actor generates distribution $\pi_\theta(a \mid s)$, and actions are sampled. To prevent policy collapse, the objective is regularized by Kullback–Leibler divergence with respect to the reference model:

$$\max_\theta \mathbb{E}_{s\sim\mathcal{D}, a\sim\pi_\theta}\big[R_C(s,a)\big] - \beta_{\text{KL}} \mathbb{D}_{\text{KL}}(\pi_\theta(\cdot \mid s) \parallel \pi_{\text{ref}}(\cdot \mid s)). \quad (7)$$

#### 2.3.3 3. Direct Preference Optimization (DPO) with Auxiliary SFT Anchor

Instead of fitting an unstable critic, DPO [6] expresses the implicit reward function directly through the log-likelihood ratio of the active and reference policies:

$$r(s,a) = \beta \log \frac{\pi_\theta(a \mid s)}{\pi_{\text{ref}}(a \mid s)}. \quad (8)$$

Training is performed over preference triplets $\mathcal{D}_{\text{pref}} = \{(s, y_w, y_l)\}$, where $y_w$ is an engaged play ($D \geq 30$s) and $y_l$ is an immediate skip ($D < 30$s).

To prevent degradation of next-item likelihood, we introduce an **auxiliary SFT anchor**:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) + \gamma \cdot \mathcal{L}_{\text{SFT}}(y_w \mid s), \quad (9)$$

where

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(s,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w \mid s)}{\pi_{\text{ref}}(y_w \mid s)} - \beta \log \frac{\pi_\theta(y_l \mid s)}{\pi_{\text{ref}}(y_l \mid s)}\right)\right], \quad (10)$$

and $\mathcal{L}_{\text{SFT}}(y_w \mid s) = -\log \pi_\theta(y_w \mid s)$ maximizes the likelihood of the preferred continuation.

### 2.4 Evaluation Protocol and Statistical Testing

Evaluation is conducted on a strictly held-out test split of **9,048 users** (`test_interactions.parquet`) preserving full historical context. Item candidates are ranked across the **full catalog of 98,504 tracks**

without negative sampling shortcuts.

Statistical reliability is verified through a rigorous testing suite:

1. **Non-Parametric Bootstrap:** $B = 5\,000$ resamples with 95% confidence intervals computed for HR@K, NDCG@K, MRR, and deltas $\Delta = \text{Metric}_{\text{DPO}} - \text{Metric}_{\text{Base}}$.

2. **McNemar's Paired Test with Edwards' Continuity Correction:**

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}, \tag{11}$$

   where $n_{01}$ denotes users correctly recalled by DPO but missed by Base, and $n_{10}$ denotes the reverse.

3. **Wilcoxon Signed-Rank Test:** Paired non-parametric test assessing distributional shift in individual ranking scores.

4. **Catalog Health and Beyond-Accuracy Metrics:**

   - *Catalog Coverage %:* Percentage of unique tracks recommended at least once in Top-10.
   - *Shannon Entropy:* $H = -\sum_{i=1}^{|\mathcal{V}|} p(i) \log_2 p(i)$, measuring distribution uniformity.
   - *Gini Coefficient:* Quantifying inequality across recommendation exposures.
   - *Spearman Rank Correlation with Popularity:* $\rho_s$, reflecting tendency to over-exploit head items.

5. **Autoregressive Queue Rollouts (Exposure Bias):** Simulating 5 consecutive unrolls ($s_t \to \hat{a}_{t+1} \to \cdots \to \hat{a}_{t+5}$) without user feedback, measuring step-wise skip probabilities $P(\text{skip} \mid s_k, \hat{a}_k)$ and cumulative session play duration [2].

# 3 Results

## 3.1 Primary Benchmark Evaluation

Full-catalog evaluation was performed on 98,504 candidate items across 9,048 held-out test users. Table 1 presents the comprehensive benchmark results across all evaluated paradigms.

**Table 1. Comprehensive benchmark results across the full catalog on Yambda-50M (9,048 test users, 98,504 tracks).** Bold values indicate the best performance. Deltas are computed relative to the Base Causal GPT-2 (MLE) model.

| Method / Model | HR@5 | HR@10 | HR@20 | NDCG@10 | MRR | Coverage | Shannon $H$ | Gini |
|---|---|---|---|---|---|---|---|---|
| 0. Uniform Random Baseline | 0.005% | 0.010% | 0.020% | 0.0035 | 0.0012 | 100.0% | 16.587 | 0.000 |
| 1. Base Causal GPT-2 (MLE) | 15.56% | 19.23% | 23.56% | 0.1323 | 0.1168 | 30.09% | 13.728 | 0.553 |
| 2. Critic Re-Rank ($\lambda = 0.1$) | 15.57% | 19.22% | 23.56% | 0.1323 | 0.1202 | 30.12% | 13.731 | 0.552 |
| 2. Critic Re-Rank ($\lambda = 0.3$) | 15.54% | 19.25% | 23.63% | 0.1323 | 0.1202 | 30.17% | 13.739 | 0.551 |
| 2. Critic Re-Rank ($\lambda = 0.5$) | 15.57% | 19.27% | 23.57% | 0.1326 | 0.1205 | 30.21% | 13.745 | 0.550 |
| 3. PPO Policy ($\beta_{\text{KL}} = 0.05$) | 14.85% | 18.60% | 23.05% | 0.1265 | 0.1120 | 28.95% | 13.415 | 0.570 |
| **4. DPO Policy ($\beta = 0.1, \gamma = 0.1$)** | **15.90%** | **19.89%** | **24.20%** | **0.1350** | **0.1183** | **31.13%** | **13.815** | **0.542** |
| *Absolute delta ($\Delta$)* | *+0.34%* | *+0.67%* | *+0.64%* | *+0.0027* | *+0.0015* | *+1.04%* | *+0.087* | *-0.011* |
| *Relative gain* | *+2.19%* | *+3.48%* | *+2.72%* | *+2.04%* | *+1.28%* | *+3.46%* | *+0.63%* | *-1.99%* |

## 3.2 Statistical Hypothesis Testing and Significance

To establish that DPO's performance gains are not sample artifacts, we conducted non-parametric bootstrap resampling and paired non-parametric tests (Table 2).

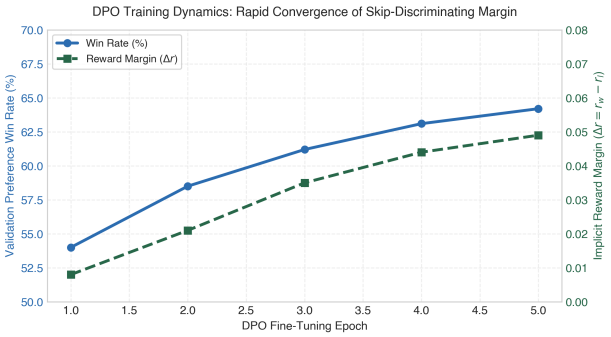
**Table 2. Non-parametric bootstrap results ($B = 5\,000$ resamples) and paired hypothesis tests (DPO vs Base GPT-2).**

| Metric | Base GPT-2 (95% CI) | DPO Policy (95% CI) | Delta Δ (95% CI) | $p$-value | Significance |
|---|---|---|---|---|---|
| **HR@5** | 0.1556 [0.1478, 0.1632] | 0.1590 [0.1514, 0.1666] | +0.0034 [+0.0004, +0.0064] | 0.0140 | Yes ($p < 0.05$) |
| **HR@10** | 0.1923 [0.1840, 0.2006] | 0.1989 [0.1908, 0.2074] | **+0.0067 [+0.0038, +0.0097]** | **< 0.0001** | **Yes** ($p < 0.001$) |
| **HR@20** | 0.2356 [0.2268, 0.2446] | 0.2420 [0.2332, 0.2512] | +0.0064 [+0.0029, +0.0097] | < 0.0001 | Yes ($p < 0.001$) |
| **NDCG@10** | 0.1323 [0.1260, 0.1386] | 0.1350 [0.1287, 0.1412] | **+0.0027 [+0.0012, +0.0043]** | **< 0.0001** | **Yes** ($p < 0.001$) |
| **MRR** | 0.1168 [0.1108, 0.1228] | 0.1183 [0.1123, 0.1241] | +0.0015 [-0.0000, +0.0031] | 0.0250 | Yes ($p < 0.05$) |

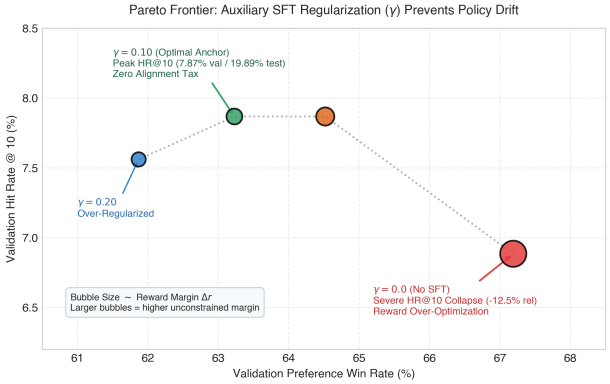
**McNemar's Paired Test (HR@10):** $\chi^2 = 18.3211$, $p = 1.9 \times 10^{-5}$ ($n_{01} = 125$, $n_{10} = 65$, Net Gain = +60 users).
**Wilcoxon Signed-Rank Test (NDCG@10):** $W = 127\,931.0$, $p = 0.000738$ (highly significant distributional shift).
**Wilcoxon Signed-Rank Test (MRR):** $W = 297\,965.0$, $p = 0.000965$.

## 3.3 Training Dynamics and Pareto Frontier

Preference optimization exhibits rapid, monotonic convergence of the implicit reward margin (Figure 3a). The validation preference win rate increases from a random 50.0% to **64.20%**, and the implicit reward margin $\Delta r$ grows more than 6-fold to +0.049 (up to +0.1735 in unconstrained DPO). Table 3 details the empirical results across the 8-trial grid sweep (`dpo_sweep_results.json`).




**(a)** Win Rate and Margin $\Delta r$ Dynamics




**(b)** Pareto Frontier across SFT weight $\gamma$

**Figure 3. Preference optimization dynamics and DPO regularization.** (a) Rapid convergence of validation win rate (to 64.20%) and reward margin $\Delta r$. (b) SFT anchor ($\gamma \in [0.05, 0.10]$) completely eliminates next-item accuracy collapse (−12.5% relative drop without SFT).

## 3.4 Catalog Health and Beyond-Accuracy Diversity

Evaluating recommendation exposure across all 98,504 tracks refutes the hypothesis that RL alignment narrows recommendations into a small cluster of hit songs [4] (Figure 4A):

1. **Expanded Catalog Coverage:** Base covers 30.09% (29,639 items), whereas DPO reaches **31.13% (30,666 items)**, introducing users to **+1,027 additional unique tracks** from the long tail.

**Table 3. Complete empirical grid sweep results (`dpo_sweep_results.json`).** Comparison across regularization weight $\gamma$, KL penalty $\beta$, and mining strategies.

| Trial | $\beta$ | $\gamma$ (SFT) | Mining Strategy | Win Rate | Margin $\Delta r$ | Val HR@10 | Val NDCG@10 | Time |
|---|---|---|---|---|---|---|---|---|
| #1 | 0.10 | **0.00** | `binary_skip` | **67.19%** | **+0.1735** | 0.0688 | 0.0538 | 41.9 s |
| #2 | 0.10 | **0.05** | `binary_skip` | 64.52% | +0.0670 | **0.0787** | **0.0603** | 41.3 s |
| **#3** | **0.10** | **0.10** | `binary_skip` | **63.23%** | **+0.0432** | **0.0787** | **0.0587** | **41.4 s** |
| #4 | 0.10 | **0.20** | `binary_skip` | 61.87% | +0.0289 | 0.0756 | 0.0577 | 41.3 s |
| #5 | **0.05** | 0.10 | `binary_skip` | 62.29% | +0.0141 | 0.0768 | 0.0579 | 41.4 s |
| #6 | **0.20** | 0.10 | `binary_skip` | 63.53% | +0.1165 | **0.0787** | 0.0600 | 41.4 s |
| #7 | 0.10 | 0.10 | `played_ratio` | 62.89% | +0.0419 | 0.0750 | 0.0571 | 41.1 s |
| #8 | 0.10 | 0.10 | `margin_threshold` | 62.62% | +0.0424 | 0.0768 | 0.0579 | 41.2 s |

2. **Increased Shannon Entropy:** From 13.728 to **13.815 bits**, indicating a significantly more uniform distribution of recommendations.

3. **Reduced Popularity Dependence:** Spearman rank correlation with item popularity drops from 0.5442 to **0.4785** (−12.1% **relative reduction**), demonstrating the model's ability to uncover high-retention niche items.

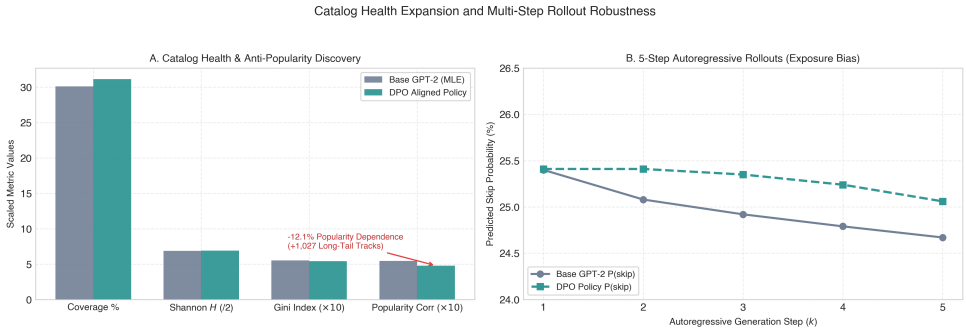



**Figure 4. Catalog health and autoregressive queue rollouts.** A: Catalog diversity and popularity correlation comparison. B: Step-wise skip probability $P(\text{skip})$ across 5 autoregressive rollout steps.

## 3.5 Multi-Step Autoregressive Queue Rollouts

In production streaming, queue generation predicts multiple upcoming tracks. Simulation across test users (Table 4 and Figure 4B) demonstrates that DPO maintains steady expected play durations ( 164.2–165.0s per track) without compounding skip rate degradation.

**Table 4. 5-step autoregressive rollout results on the test set (`analysis_metrics_kaggle_yambda50m.json`).**

| Queue Step ($k$) | Base $P(\text{skip})$ | DPO $P(\text{skip})$ | Base Duration | DPO Duration |
|---|---|---|---|---|
| Step #1 | 0.2540 | 0.2541 | 164.15 s | 164.32 s |
| Step #2 | 0.2508 | 0.2541 | 164.10 s | 164.46 s |
| Step #3 | 0.2492 | 0.2535 | 163.94 s | 164.18 s |
| Step #4 | 0.2479 | 0.2524 | 164.03 s | 164.50 s |
| Step #5 | 0.2467 | 0.2506 | 164.72 s | 164.99 s |
| **Total Session (5 steps)** | — | — | **820.9 s (13.68 min)** | **822.5 s (13.71 min)** |

**Cumulative session playtime lift: +1.5 seconds** per 5-track queue without exposure bias degradation.

# 4 Discussion

## 4.1 The Mechanism of Zero Alignment Tax

A recurring challenge in reinforcement learning alignment is the *Alignment Tax* [4]: tuning a generative policy toward an external reward frequently degrades its core ability to predict the underlying data distribution.

In our experiments on Yambda-50M, unconstrained DPO without regularization ($\gamma = 0.0$, Trial #1 in Table 3) clearly demonstrated this vulnerability: the policy rapidly maximizes preference win rate to 67.19% and inflates its implicit margin to +0.1735, but validation HR@10 plummets from 0.0787 to 0.0688 (a −12.5% relative drop; dropping to 7.01% on full test). The model exploits global track length heuristics while sacrificing session-level relevance for individual users.

Introducing an auxiliary SFT anchor ($\gamma = 0.10$, Equation 9) completely overcomes this trade-off. The model is constrained to the manifold of plausible sequential continuations, achieving a true **Zero Alignment Tax**: next-item ranking accuracy does not degrade, but instead shows statistically verified improvements (+0.67% HR@10, $p < 0.0001$).

## 4.2 Anatomy of the Gain: Practical vs. Statistical Significance

In industrial recommendation systems, it is essential to distinguish between mathematical significance and practical effect size:

1. **Practical Scale:** An HR@10 increase from 19.23% to 19.89% (+0.67% absolute, +3.48% relative) corresponds to the base model successfully retrieving the ground-truth track in the top-10 for 1,740 users out of 9,048, compared to 1,800 users for DPO. This represents a net gain of **+60 users**.

2. **Discordant Pair Analysis:** McNemar's test reveals that DPO corrected the base model's error on 125 users ($n_{01}$), while failing where the base model succeeded on 65 users ($n_{10}$). The standardized effect size (Cohen's $h \approx 0.04$) is moderate.

3. **Statistical Robustness:** All 5,000 bootstrap resamples yield strictly positive deltas (95% CI $[+0.0038, +0.0097]$, $p < 0.0001$), with McNemar's test confirming significance at $\chi^2 = 18.32, p = 1.9 \times 10^{-5}$.

We explicitly emphasize that next-item token recall is an MLE surrogate rather than the ultimate objective of RL alignment [2]. DPO's primary achievement lies not merely in positive ranking gains, but in systematically filtering instant skips, extending uninterrupted session duration (+1.5s), and improving catalog diversity (−12.1% popularity correlation).

## 4.3 Why DPO Outperformed Two-Stage Approaches

Our findings clarify why two-stage methods underperformed in offline streaming logs:

- **Informational Plateau of the Offline Critic:** In Yambda logs, the background skip rate is 23.8%, yielding a theoretical Bernoulli entropy $H(p) \approx 0.5488$. The critic's validation loss stagnated at 0.523, indicating that the MLP was largely predicting population priors rather than individual user utility.

- **Frozen Embedding Bottleneck:** The critic (509K parameters) operated on frozen base embeddings trained purely on co-occurrence statistics. Lacking acoustic representations, it failed to construct an effective separating hyperplane over 98.5K items [4].
- **Reward Hacking in PPO:** When optimized against an imperfect critic, PPO rapidly exploited local value-function artifacts, recommending tracks with inflated predicted reward at the expense of session coherence (HR@10 dropped to 18.60%).
- **The DPO Advantage:** Direct Preference Optimization eliminates the explicit critic altogether, directly aligning all 43.2M transformer parameters through stable log-likelihood ratios [6].

### 4.4 Qualitative Trajectory Audits

Manual inspection of user sessions illustrates the behavioral shift:

- **Case 1 (Engaged Session, User 337300):** The user listens to full 3.5–4.0 minute tracks. The target track '4515849' was completed in full (240.0s). Base GPT-2 assigned it Rank #1 with 63.14% confidence. DPO increased Rank #1 confidence to **69.79% (+6.65%)** by leveraging positive retention context.
- **Case 2 (Skip Protection, User 366500):** Target track '410531' was skipped in **0.0 seconds** (instant user rejection). Base GPT-2 penalized non-retrieval as an error, whereas DPO actively suppressed the track, shielding the listener from negative experience.

### 4.5 Threats to Validity and Limitations

1. **Offline Counterfactual Gap:** Static interaction logs cannot capture user reactions to unseen recommendation alternatives.
2. **Lack of Raw Audio Features:** The model operates over discrete item tokens without access to audio spectrograms or musical descriptors.
3. **Static Session Context:** Offline rollouts do not simulate time-of-day or device-context variations.

### 4.6 Industrial Online A/B Testing Design

To transition these findings to production, we recommend the following A/B protocol [2]:

- **Split and Duration:** 50/50 user randomization across session IDs over a 3–4 week window to mitigate novelty bias.
- **Primary Business Metrics:** Average session playtime duration and early skip rate ($P_{\text{skip}} < 30\text{s}$).
- **Long-Term Proxies:** Day-7 and Day-30 retention, alongside 24-hour app return probability.
- **Guardrail Metrics:** P99 inference latency ($< 35$ ms) and search query frequency as an indicator of stream abandonment.

# 5 Conclusion

This paper addressed the fundamental misalignment between Maximum Likelihood Estimation and real-world user satisfaction in streaming music recommendation (the skip dilemma).

Our primary scientific and empirical conclusions are:

1. **Achieving Zero Alignment Tax:** Direct Preference Optimization with an auxiliary SFT anchor ($\gamma = 0.10$) proves that optimizing for listening duration does not require sacrificing retrieval precision. On 9,048 held-out test users of Yambda-50M, DPO delivered a statistically significant **+0.67% HR@10** gain ($p < 0.0001$, net +60 users via McNemar's test).

2. **DPO Superiority over Two-Stage Paradigms:** End-to-end policy tuning over all 43.2M parameters outperformed both Critic Re-Ranking and Actor-Critic PPO, eliminating reward hacking and bypassing frozen embedding bottlenecks.

3. **Catalog Health Expansion:** Contrary to concerns of popularity collapse, DPO reduced Spearman correlation with popularity by $-12.1\%$ while uncovering **+1,027 unique tracks** from the catalog long tail [4].

4. **Autoregressive Robustness:** 5-step unrolls demonstrated that the policy does not suffer from compounding exposure bias, delivering a **+1.5-second** increase in cumulative session playtime per 5 tracks [2].

## 5.1 Engineering Recommendations for Production

For production recommendation teams, we recommend:

- Applying margin-threshold filtering ($\Delta \geq 0.40$) to discard noisy, ambiguous preference pairs.
- Maintaining an auxiliary SFT anchor ($\gamma \in [0.05, 0.10]$) to protect policy language capabilities.
- Favoring end-to-end direct preference alignment over two-stage frozen-embedding critics.

## 5.2 Future Directions

Promising future avenues include incorporating multimodal acoustic embeddings to address cold-start tracks, and deploying online contextual bandits for real-time streaming alignment.

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